Aayush Kumar

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Summary

Computer Science Engineering student specializing in Artificial Intelligence and Machine Learning with strong programming skills in Python, Java, SQL, and Data Structures & Algorithms. Experienced in developing database-driven applications and cloud-based solutions using Azure and Oracle Cloud. Passionate about software engineering, AI, and backend development with hands-on experience in Git, Docker, Linux, and database design.

Education

Parul Institute Of Engineering And Technology **August 2024 – June 2028**
Bachelor of Technology in Computer Science Engineering | CGPA: 7.5 *Vadodara, India*

Maharashtra State Board (HSC) **June 2024**
Science (PCM) *Mumbai, India*

Projects

BST Visualizer | *Python, Data Structures, Algorithms* **December 2025**

- Built a Binary Search Tree visualizer implementing insert, delete, and search operations
- Implemented DFS and BFS traversals including Inorder, Preorder, Postorder, and Level Order
- Designed modular Python architecture with automated testing
- Distributed the application as a standalone executable using PyInstaller

OIL India Limited SIF Precursor Intelligence System | *Python, AI/NLP, Machine Learning* **October 2026**

- Developed an NLP-based system to identify Serious Injury and Fatality (SIF) precursors in safety reports
- Mapped reports to IOGP Life-Saving Rules and identified recurring safety risk patterns
- Deployed the application on Render for web-based access

Technical Skills

Languages: Python, Java, C, SQL, JavaScript
Web Technologies: HTML, CSS
Cloud/Platforms: Microsoft Azure, Oracle Cloud Infrastructure
Databases: MySQL, MongoDB
Developer Tools: Git, GitHub, Docker, Linux, VS Code
Libraries/Frameworks: NumPy, MediaPipe

Certifications

- **AWS Cloud Practitioner Essentials** — Amazon Web Services (AWS), 2026
- **Oracle Cloud Infrastructure Foundations Associate** — Oracle, 2026
- **Agents and Workflows** — OpenAI Academy, 2026
- **Claude Code in Action** — Anthropic, 2026
- **Python (Basic)** — HackerRank, 2026

Achievements & Leadership

- **Smart India Hackathon (SIH) 2026 – Certificate of Participation**
- Participated in EY Techathon 6.0, collaborating on innovative software solutions.